

- Most commonly, chemical reactions are shown in the form of a chemical equation (reaction equation)
 - Summarizes reactants and products, representing them using their formulas
 - Ex: Reactant + reactant $\rightarrow$ product + product
 - Reactants on left side, products on right; $\rightarrow$ represents the reaction happening (“yields”)
 - There are additional symbols for things like the states of matter of materials, or if there were other conditions in the reaction environment

- Initial reaction equation = skeleton equation
 - Shows the formulas of the elements/compounds involved in the reaction
 - Shows which are reactants and which are products
 - Good for generally conveying what the reaction is
 - Doesn't take into account law of conservation of mass
- Show law of conservation of mass by balancing the equation!
 - We can't change the formula of compounds themselves (changes identity of reaction!), but can alter the ratio of compounds/elements to each other

- You might be wondering why oxygen written as O_2 in combustion equation on board
- Most elements, in their elemental state, we say would exist as ~individual atoms
- However, there are 7 very common elements whose atoms are more stable, and therefore naturally exist, in pairs of those atoms - called **diatomic elements**
- H_2 N_2 O_2 F_2 Cl_2 Br_2 I_2
- There are a couple of mnemonic devices to help you remember
 - **Have No Fear Of Ice Cold Beer**

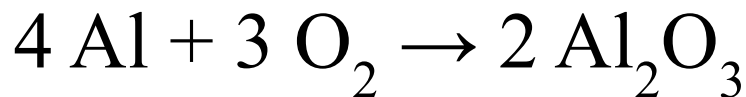
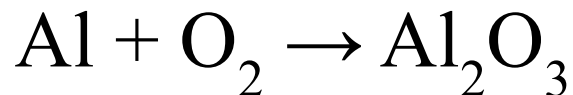
Group →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Period ↓	1 1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	* 71 Lu	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	* 103 Lr	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			* 57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb		
			* 89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No		

Go to 7, make a 7; missing 1, go to 1

- To get a balanced chemical equation, we've established the process so far as this:
 - Step 1: Balance skeleton equation by adding coefficients in front of compounds/elements OR determine that skeleton equation is already balanced
- We're going to complicate this a little bit by adding another way that we can represent a chemical reaction
 - Word equation = skeleton equation, but with names of compounds as opposed to formulas
 - Ex: potassium chlorate $\rightarrow$ potassium chloride + oxygen

- Word equation to balanced equation steps:
- 1. Use names to write correct formulas for compounds (making skeleton equation)
- 2. Balance skeleton equation by adding coefficients
OR determine that skeleton equation is already balanced

Ex: aluminum + oxygen $\rightarrow$ aluminum oxide



(don't have to rewrite equation...)